



Residual compressive strength of concrete after exposure to high temperatures: A review and probabilistic models

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ARTICLE INFO

Keywords:

Concrete
Post-fire
Residual compressive strength
Dataset
Probabilistic model

ABSTRACT

The mechanical properties of concrete degrade after exposure to high temperatures during a fire event. Proper assessment of damage, determination of the remaining capacity of the structure, and recommendations for repair actions require the quantification of the reduction in the compressive strength of concrete after fire. This paper collects and analyzes the existing data on residual compressive strength of normal strength concrete in the literature. The analyses aim to identify the influential attributes that could be used to reduce scatter in the data. Parameters related to aggregate type, concrete mix, and testing protocol are studied. Subsets of the dataset for concrete with siliceous and calcareous aggregates are created. Data for siliceous concrete is further categorized based on the retaining time of the maximum temperature during the test. The subsets are used to propose probabilistic models for the residual compressive strength of concrete with siliceous and calcareous aggregates. The models follow the Weibull distribution and are continuous functions that can be implemented in analytical and computational frameworks. The medians of the proposed model and data are close to the Eurocode reduction factors for the siliceous concrete but fall below the Eurocode reduction factors for concrete with calcareous aggregate.

1. Introduction

Fire is a major societal hazard with the potential to disrupt infrastructure systems and drain valuable resources from a community. Extreme fire events, such as the 1999 Mont Blanc Tunnel fire and the 2017 London Grenfell Tower fire, can lead to fatalities and significant damage. The dollar cost of structure fires increases significantly when accounting for indirect losses and expenditures, with estimates of around 1% of the GDP in developed countries [25,77]. Concrete structures typically behave well at high temperatures and do not collapse when subjected to moderate fires. The superior performance of concrete structures under fire, compared to other materials, relates to the low thermal conductivity of the material and the large mass of the structural components. Nonetheless, these structures experience damage and require repair to resume function after a fire.

The available guidelines on damage assessment of concrete structures after a fire are limited [2,3,23,24,32], and the evaluation is performed mostly on an ad hoc basis. Post-fire damage and repair

classifications in the current guidelines rely on visual inspection, non-destructive testing, and sampling of material for laboratory testing. Meanwhile, the information from visual inspections and non-destructive testing can be supplemented with results from advanced modeling and an analysis of the structure subjected to fire to guide the damage assessment procedure and develop proper repair strategies. Aside from the fire scenario, the models require input on the thermal and mechanical properties of materials at high temperatures. Given the large levels of uncertainty in fire events and the properties of materials at high temperatures, a proper assessment of said performance involves stochastic evaluation of the structural response. Previous work studied and quantified the variability in the mechanical properties of steel and concrete at high temperatures [30,50], but additional research is needed to quantify the post-fire mechanical properties of concrete for damage and recovery assessment.

There is a relatively large number of experimental studies on the residual compressive strength of concrete after exposure to high temperatures. Babalola et al. [11] conducted a review study on the residual

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<https://doi.org/10.1016/j.firesaf.2022.103698>

Received 6 May 2022; Received in revised form 23 October 2022; Accepted 29 October 2022

Available online 3 November 2022

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mechanical properties of concrete exposed to elevated temperatures, focusing on the effects of natural and recycled aggregates, mineral admixtures, and fibers. The study provided some noteworthy findings for improving the performance of concrete after exposure to elevated temperatures; however, estimating the residual strength of concrete remains a challenge due to the large scatter in the data. The large variability in existing datasets is related to the inherent uncertainty in the material behavior in addition to variations in various attributes of the experiments, e.g., a lack of consistency in testing protocols.

The objective of this study is to provide a comprehensive review of experimental investigations on the residual compressive strength of concrete after exposure to high temperatures. The existing data are analyzed to identify attributes that could affect the dispersion in residual strength, such as the aggregate type, concrete mix, and testing protocol. Categories with limited data points that require further research are identified. The refined datasets are then used to develop probabilistic models for the normalized residual compressive strength of concrete. The proposed models have simple mathematical forms that can be applied in analytical and computational frameworks. Both data and proposed models are compared with the Eurocode [17,18] prescribed values for the residual strength of concrete after fire and the suitability of the Eurocode reduction factors is discussed.

2. Methodology

The content analysis approach proposed by Mayring [44] is applied to collect and review the available literature on residual compressive strength of concrete after exposure to high temperatures. The review is carried out systematically, following a four-step process:

1. Material collection: comprises a well-organized search and delimitation of articles.
2. Descriptive analysis: identifies the general features of the literature under review.
3. Category selection: builds a categorization framework based on relevant dimensions. For this study, attributes of the data that could influence the outcome are identified.
4. Material evaluation: reviews and interprets articles using the given categorization framework.

The first three steps are discussed in the following subsections. The final step is explored in greater detail along with the analysis of collected data on the residual compressive strength of concrete.

3. Material collection

The following criteria were considered when conducting the literature review: (1) papers that provided experimental data on the residual compressive strength of concrete subjected to high temperatures were collected, (2) papers that included experimental data on compressive strength of concrete at high temperatures (i.e., measurements done while specimens were hot) were excluded, and (3) published journals papers for a time span of 30 years (from 1990 to 2020) were included. Considering these criteria, a total of 60 articles containing 1400 data points on the residual compressive strength of concrete as a function of temperature were identified. Further, only the data points for concrete with a compressive strength of less than 55 MPa at room temperature, i.e., normal strength concrete (NSC), were kept in the dataset. Thus, 53 articles with a total of 1240 (out of 1400) data points were selected for review and analysis.

3.1. Descriptive analysis

Fig. 1 presents the distribution of published experimental data from 1990 to 2020. The number of experimental studies on the residual compressive strength of concrete subjected to high temperatures has an

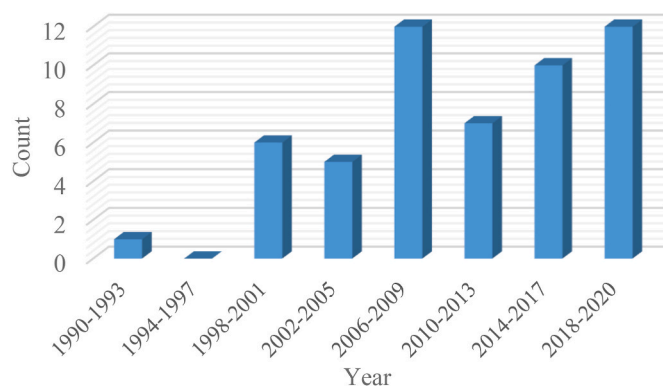


Fig. 1. Distribution of reference papers by year.

increasing trend over time.

The distribution of specimen types in the examined literature is shown in Fig. 2. There is a relatively uniform distribution between cube versus cylinder specimens. In most of the European studies, the compressive strength is measured using cubic specimens, whereas cylinders are used in the United States and Australia. Pacheco et al. [46] proposed a cube to cylinder conversion factor of 0.81 for normal aggregates and 0.77 for recycled aggregates. The conversion factors were applied to cubic samples before grouping the data based on concrete strength (normal vs. high strength).

3.2. Category selection

The collected literature included variations in a series of attributes, including the aggregate type, concrete mix, and testing protocol, when investigating the residual compressive strength of concrete as a function of temperature. These attributes are listed in Fig. 3 and their influence on the residual strength of concrete is studied in this paper.

About 80% of concrete is made up of coarse aggregate and the aggregate type has an effect on the residual compressive strength of concrete that is exposed to high temperatures [16]. The most often used categories for aggregate types are siliceous and calcareous aggregates. Calcareous aggregates are sedimentary rocks that are primarily composed of calcium carbonate (CaCO_3), such as limestone or dolomite, whereas siliceous aggregates are mainly composed of silica (SiO_2 , i.e., combination of silicon and oxygen) and silicates (combination of silicon, oxygen, and other elements). Thermal incompatibility between the aggregate and cement paste causes degradation in mechanical properties of concrete at high temperatures. The higher the aggregate's coefficient of thermal expansion, the more damage the concrete experiences when exposed to high temperatures [17].

The rest of the considered parameters are related to the concrete mix and testing protocol. Concrete mix includes the water/cement (w/c) ratio and inclusion of any fibers and/or supplementary cementitious materials (SCM). The testing protocol refers to the heating rate, retaining time, cooling rate, and cooling method (e.g., air or water).

Table 1 summarizes the type and range of parameters in the collected

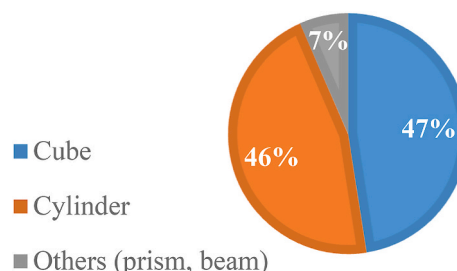


Fig. 2. Distribution of specimen type in the collected data.

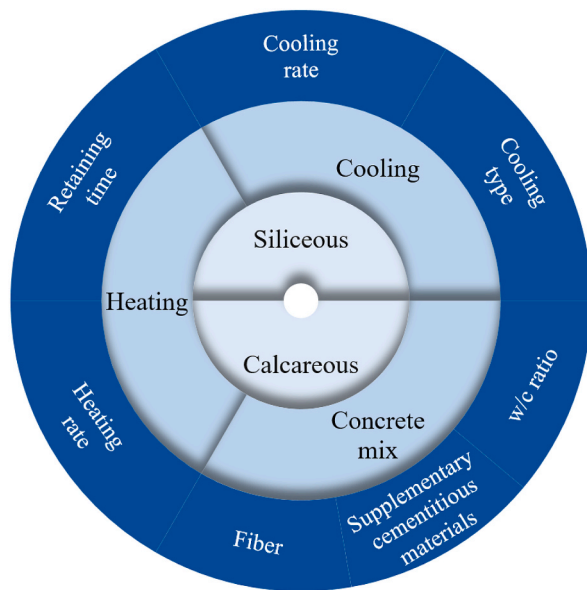


Fig. 3. Collected parameters that could influence residual compressive strength of concrete exposed to high temperatures.

Table 1
Description, variation, and range of parameters in the collected dataset.

Parameter	Unit	Range
Aggregate	–	Siliceous, calcareous
Fibers	–	Polypropylene fiber, steel fiber, basalt fiber, hybrid fiber
Supplementary cementitious material	–	Fly ash, silica fume, pozzolan, metakaolin, slag, sludge ash, husk ash, etc.
w/c or w/b ^a	–	0.28–0.77
Heating rate	°C/min	0.83–20 (Electric furnace) ISO 834 or ASTM E119 (Gas furnace)
Retaining time	hr.	1–9
Cooling rate	°C/min	0.40–3
Cooling type	–	Air, water (spray, jet, or tank)

^a The water to binder (w/b) ratio, instead of w/c ratio, was reported for mixtures containing supplementary cementitious materials (i.e., fly ash, slag, or silica fume).

dataset. Note that other factors such as the size of the specimen, concrete age at testing, moisture content, etc. could also influence the residual strength of concrete but consistent information was not available across studies in the literature, thus not included in this research.

4. Material evaluation

The following sections will categorize and analyze the dataset to identify parameters that influence the residual compressive strength of concrete. These parameters are used to define sub-groups within the dataset that perform similarly and exhibit less scatter. Observations from collected data are also compared with reduction factors from Eurocode [17,18] to assess the suitability of prescribed values by the code. Finally, analysis of data is used to identify areas that require further investigation.

4.1. Effect of coarse aggregate

The selection and characterization of the coarse aggregate likely has the most influence on the performance of concrete at high temperatures

[60]. As previously stated, siliceous and calcareous aggregates are the two major coarse aggregate types utilized in concrete mixtures. Based on the type of coarse aggregate, Eurocode 2 (EC2) [17] specifies the loss in concrete strength due to temperature and provides reduction factors for compressive strength at high temperatures. Eurocode 4 (EC4) [18] recommends an additional 10% loss of strength, with respect to the value at the maximum achieved temperature, to determine the residual strength after exposure to high temperatures.

Table 2 lists the collected articles in this research based on the aggregate type in tested specimens. The collected dataset, targeting publications post 1990, includes 1240 data points with a substantial percentage focusing on siliceous aggregate, accounting for roughly 87% of the dataset. The available data on calcareous aggregate is very limited, as shown in Fig. 4a. Considering different aggregate types, the majority of the tests were conducted at 200 °C, 400 °C, and 600 °C, as shown in Fig. 4b.

Fig. 5 presents the normalized residual compressive strength of concrete as a function of temperature and based on aggregate type. The residual compressive strength data (f'_c) were normalized using the average of values recorded at room temperature ($f'_{c,20}$). The plots include the EC2 reduction factors for the compressive strength of concrete at high temperatures as well as the EC4 recommended reduction factors for residual strength for comparison with the data. The normalized residual compressive strength for both aggregate types show large variations across different temperatures. The figures also include box plots, indicating the first quartile (Q1), third quartile (Q3), median, whiskers that span 1.5 times of the inter-quartile range, and the outliers. The median of the data for siliceous aggregate is slightly higher than the EC4 recommendation (EC.Si), whereas in the case of calcareous aggregate, the median of data falls below the EC4 recommendation (EC.Ca), noting that there is a smaller number of data points available compared to the siliceous category.

Given the need for more sustainable construction, partial or full replacement of all natural aggregates with recycled aggregates has been considered as an option, and several published papers included concrete mixes with recycled aggregates. The experiments assessed the effects of using recycled aggregate on various concrete qualities, including residual compressive strength when subjected to harsh environments and extreme events such as high temperatures. In the siliceous category, 24% of the articles (11 out of 46) included experimental data on recycled aggregate concrete, whereas only 1 out of 12 studies (i.e., 8%) in the calcareous category was concerned with the use of recycled aggregate. Most experiments looked at the effect of using different ratios of recycled aggregates within the mix. Other design criteria such as w/c ratio, additives, and testing protocol are also investigated when recycled aggregate is used. The relevant literature including specimens with recycled aggregate is listed in Table 3.

Fig. 6 shows the normalized residual strength of siliceous concrete grouped based on normal and recycled aggregates. Excluding the data with recycled aggregate from the pool does not reduce the large dispersion in the dataset. Based on the box plots and the overall trend in the data, differentiation between recycled versus normal aggregates does not have a noticeable effect on the residual compressive strength of concrete; thus, the rest of this paper will continue to group recycled and normal aggregates together.

Table 2
Distribution of articles based on aggregate type.

Aggregate type	References
Siliceous	[1,6,7,9,12–15,19–22,26,31,36,38–43,48,49,52–58,61–67, 69–76] ^a
Calcareous	[4,8–10,22,29,33,47,51,58,66,68] ^a

^a Four articles investigated both calcareous and siliceous aggregates [Arioz 2007; Sarhat and Sherwood 2013; Chen et al. 2014; Tufail et al. 2017].

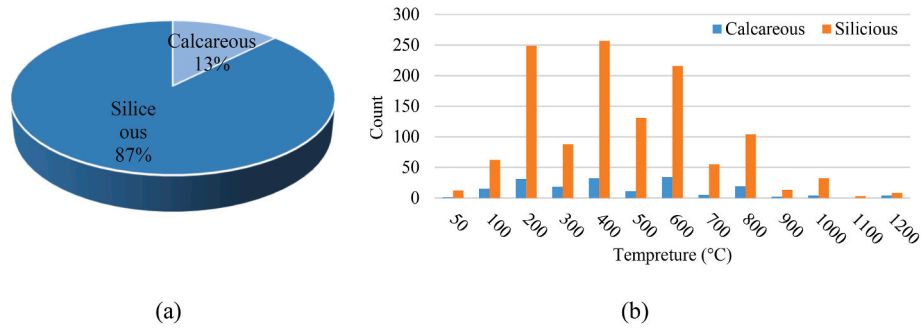


Fig. 4. Distribution of the dataset based on (a) aggregate type (b) the maximum temperature at which the test was conducted.

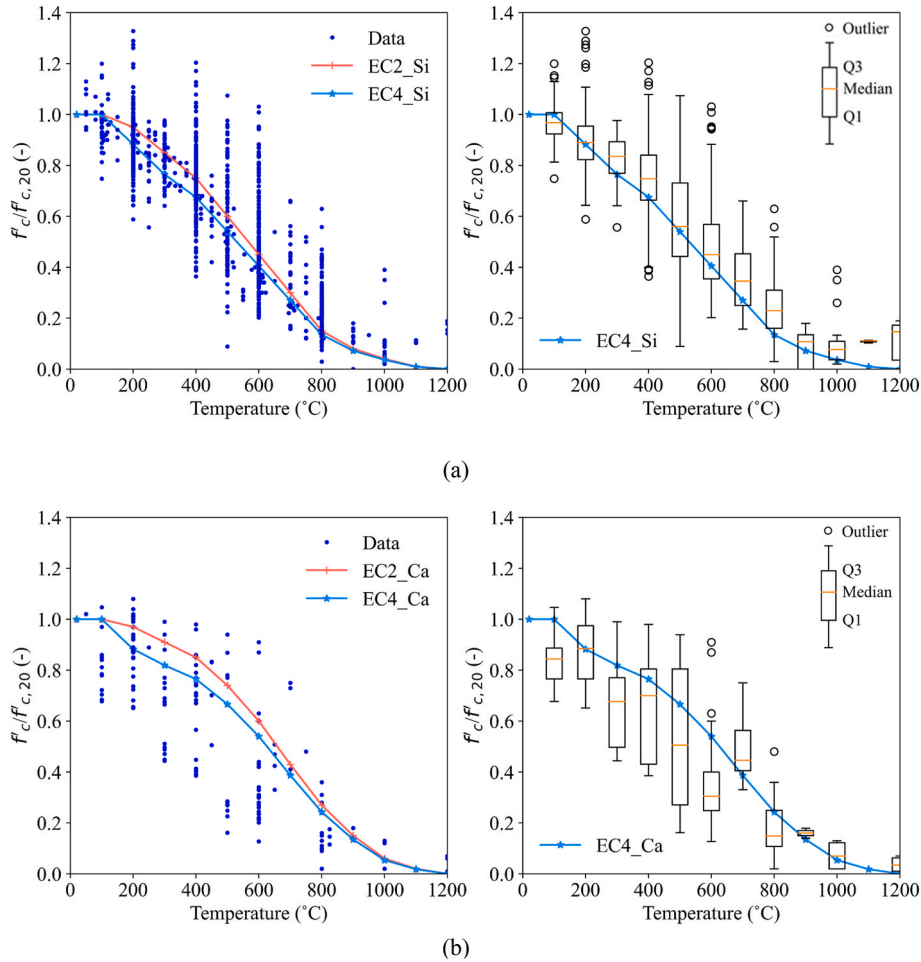


Fig. 5. Normalized residual strength of concrete as a function of temperature for experimental data, the EC2 reduction factors, and EC4 recommendation for (a) siliceous and (b) calcareous aggregate types.

Table 3

List of studies with recycled aggregate in concrete mix.

Aggregate type	Varying parameter	References
Siliceous with recycled aggregate	Ratio of recycled aggregate	[15,36,42,56,58,69,71,76]
	w/c ratio	[71,74]
	Additive	[53,56,70,73]
	Retaining time	[74]
	Cooling regime	[71]
Calcareous with recycled aggregate	Additive and cooling rate	[22]

4.2. Effect of concrete mix

The effect of different concrete mixes on the residual strength of concrete after exposure to high temperatures is the most common attribute investigated in the literature. Variations in the concrete mix can be classified based on a variety of factors, including the addition of fibers and SCM, and the w/c ratio. Table 4 lists the articles that included additives or studied the effect of w/c ratio on the residual strength of concrete. The majority of the peer-reviewed articles investigated various concrete mixes with the aim of identifying solutions to decrease spalling and cracking, as well as to improve ductility at high temperatures (e.g., [22]).

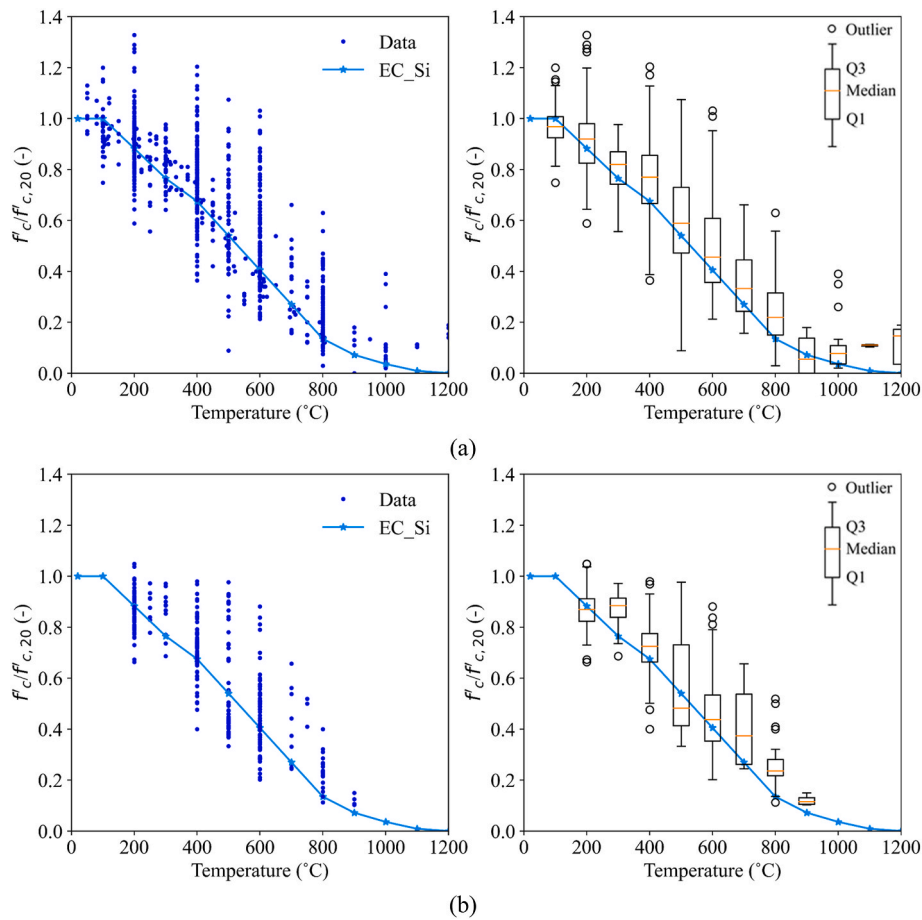


Fig. 6. Normalized residual strength of concrete with (a) normal siliceous aggregate and (b) recycled siliceous aggregate.

Table 4

List of studies investigating the effect of concrete mix on residual compressive strength of concrete after exposure to high temperatures.

Investigated parameter	List of articles	
	Siliceous	Calcareous
Fiber	[22,38,49,56,70,73]	[29,68]
SCM	[7,12,13,15,21,26,31,40,41,43,48,49,52,53,56,57,61–63,65,70,73]	[8,33]
w/c ratio	[6,9,12,19,40,65,71]	[4,9,33,47]

A number of studies investigated the effects of fibers, namely, steel, polypropylene, and hybrid (combination of steel and polypropylene), and argued that introducing such fibers may mitigate the negative effects of increasing temperatures on concrete's compressive strength (e. g., [29,38,68,70]). Fig. 7a and b shows that the residual strength of fiber-reinforced concrete with siliceous aggregates does not follow a noticeable trend, but calcareous concrete with fibers tends to have lower residual strength. It should be noted that data on calcareous concrete with fibers is from a single study [29]; thus, a general conclusion cannot be made.

Silica fume and fly ash were among the studied SCMs in the collected dataset [53]. found that adding fly ash to concrete reduced strength loss at temperatures up to 400 °C and increased strength loss at temperatures over 400 °C. Poon et al. [48] found that pozzolanic concretes incorporating fly ash had higher residual strength than pure cement concretes at temperatures below 600 °C. Concrete containing silica, on the other hand, exhibits a tendency for lower relative residual compressive strength at different temperatures [73]. Despite specific conclusions in

individual studies, Fig. 7c and d do not show a particular trend in the residual strength of concrete mixes with and without SCMs that could be used to lower the dispersion in data.

Fig. 8 Groups the collected dataset based on the w/c ratio. In general, calcareous concrete with a w/c ratio higher than 0.5 tends to retain strength after exposure to high temperatures better than concrete mixes with a lower w/c ratio, but the same conclusion does not hold true for siliceous aggregates. Therefore, the rest of this paper does not differentiate the data based on w/c ratio.

4.3. Effect of testing protocol

Nassif [45] argued that knowing the maximum exposure temperature is insufficient for estimating the postfire strength of concrete; however, no standard testing protocol exists for evaluating the effects of high temperatures on concrete, resulting in variations in conducted experiments in the literature and the potential for a large scatter in the published results [27]. The testing protocol includes the rate of heating, maximum temperature, retaining time, rate of cooling, and type of cooling. The maximum temperature for the test is typically varied to obtain reduction factors as a function of temperature. Thus, the heating rate and the retaining time are the two main variables for the heating protocol.

The heating rate is the rate at which the furnace temperature rises over time. A rapid heating rate decreases the time it takes to reach the required temperature inside a furnace, but it can lead to considerable temperature variations inside concrete specimens [37]. Sachin and Suresh [54] studied the effect of heating rate using gas and electric furnaces, and concluded that fire-exposed specimens had lower residual compressive strength than specimens heated in an electric furnace at the

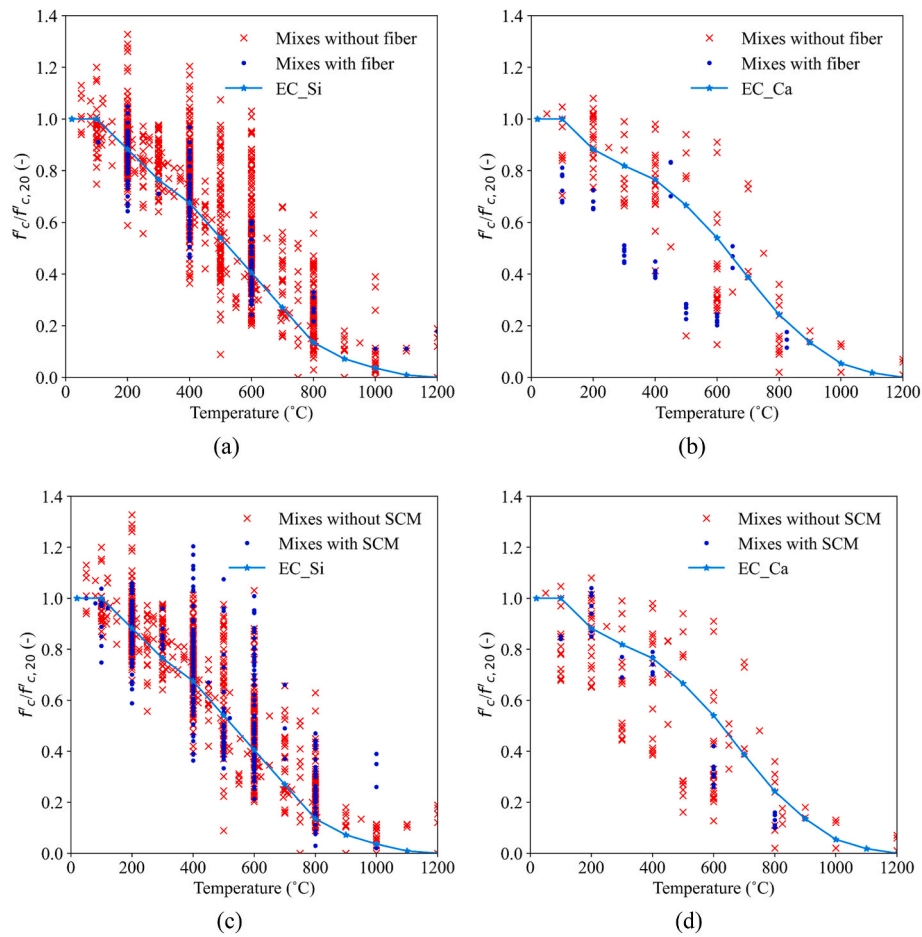


Fig. 7. Normalized residual strength of concrete categorized based on addition of fibers and SCM for (a) and (c) siliceous aggregates, and (b) and (d) calcareous aggregates.

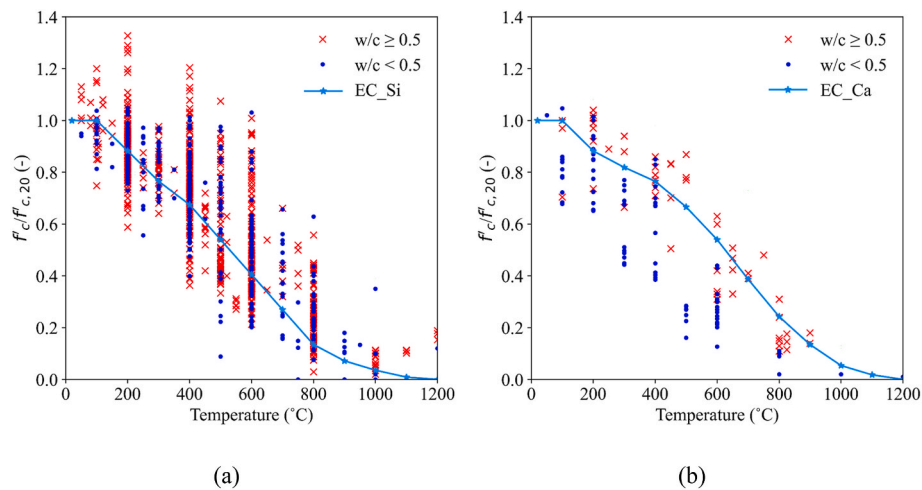


Fig. 8. Normalized residual strength of concrete with (a) siliceous and (b) calcareous aggregates categorized based on the w/c ratio.

same temperature. Gas furnaces typically follow the standard fire curve (ASTM E119 or ISO834) whereas electric furnaces have a lower heating rate compared to standard fire curves. Although the compressive strength of concrete appears to be affected by different heating rates, the majority of the reviewed articles used an electric furnace due to the safety requirements and expenses of a gas furnace. Fig. 9 groups the data based on heating rate, demonstrating that limited data is available with

the standard fire curve as the heating protocol.

Zega and Maio [74], Sakr and Hakim [55], and Zhang et al. [75] studied the effect of retaining time on the residual strength of concrete with siliceous aggregate, as the core temperature of a concrete specimen lags behind the surface temperature due to the low thermal diffusivity of concrete. The retaining time (t_R) is the duration that a specified furnace temperature is maintained once it has been reached. In general, longer

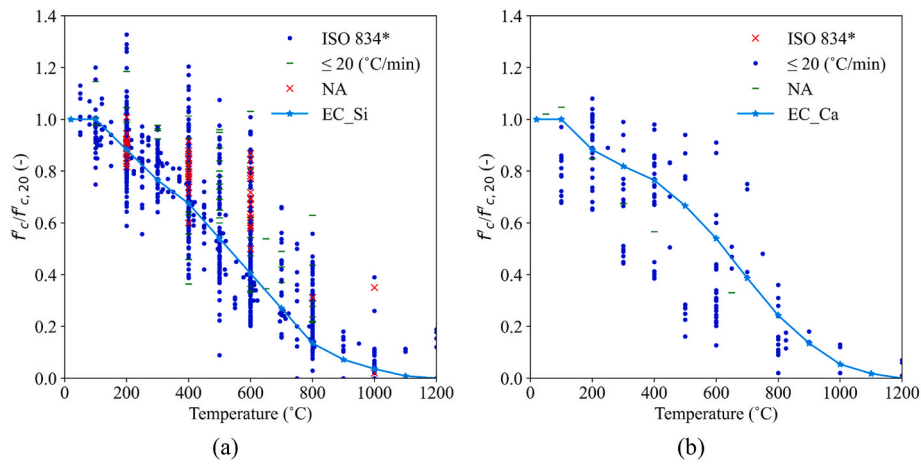


Fig. 9. Normalized residual strength of concrete with (a) siliceous and (b) calcareous aggregates categorized based on different heating rate (*or ASTM E119).

exposure times ensure a greater volume of a specimen reaches thermal equilibrium with the furnace environment, which has a considerable impact on residual compressive strength [28]. Figs. 10 and 11 group the normalized residual compressive strength data based on the retaining time of less than and larger than 2 h for siliceous and calcareous aggregates, respectively. Fig. 10 shows that introducing the retaining time to categorize the experimental data reduces the scatter in data for siliceous aggregate, especially in the range of 400–700 °C. In Fig. 10, the medians of the box plots for data with a retaining time greater than 2 h are better matched with the EC4 recommendation, whereas the medians of box plots for retaining time less than 2 h fall above the EC4

recommendation. Comparing the two categorizes emphasizes the need to allow sufficient time for the heat to reach the specimens' cores to obtain proper measurements. After investigating multiple thresholds on the dataset, the minimum 2 h of retaining time was found to be necessary for lowering the standard deviation of normalized concrete strength. Fig. 11 plots similar data but for concrete with calcareous aggregates. It is shown that data with retaining times greater than 2 h fall below the EC4 recommendation, whereas those with less than 2 h of retaining time match better with EC4. This implies that the EC4 recommendation may not be conservative for concrete with calcareous aggregates. Overall, more data is needed for concrete with calcareous

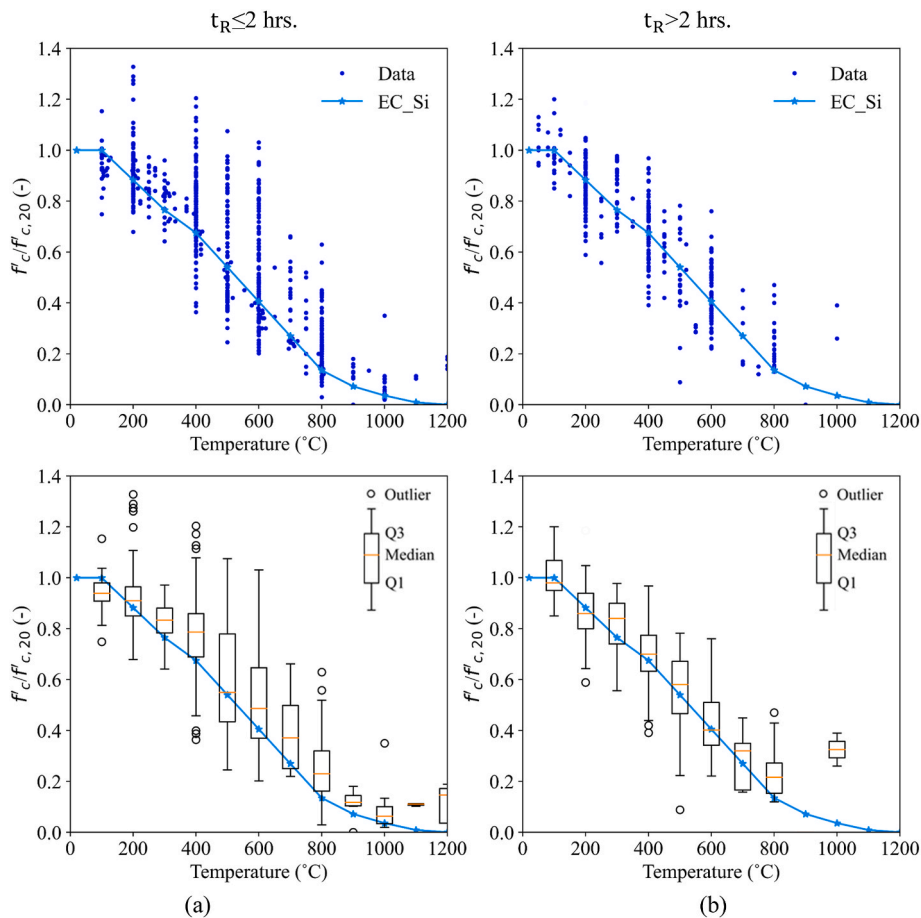


Fig. 10. Normalized residual compressive strength of concrete with siliceous aggregate and retaining time (a) less than or equal to 2 h, (b) greater than 2 h.

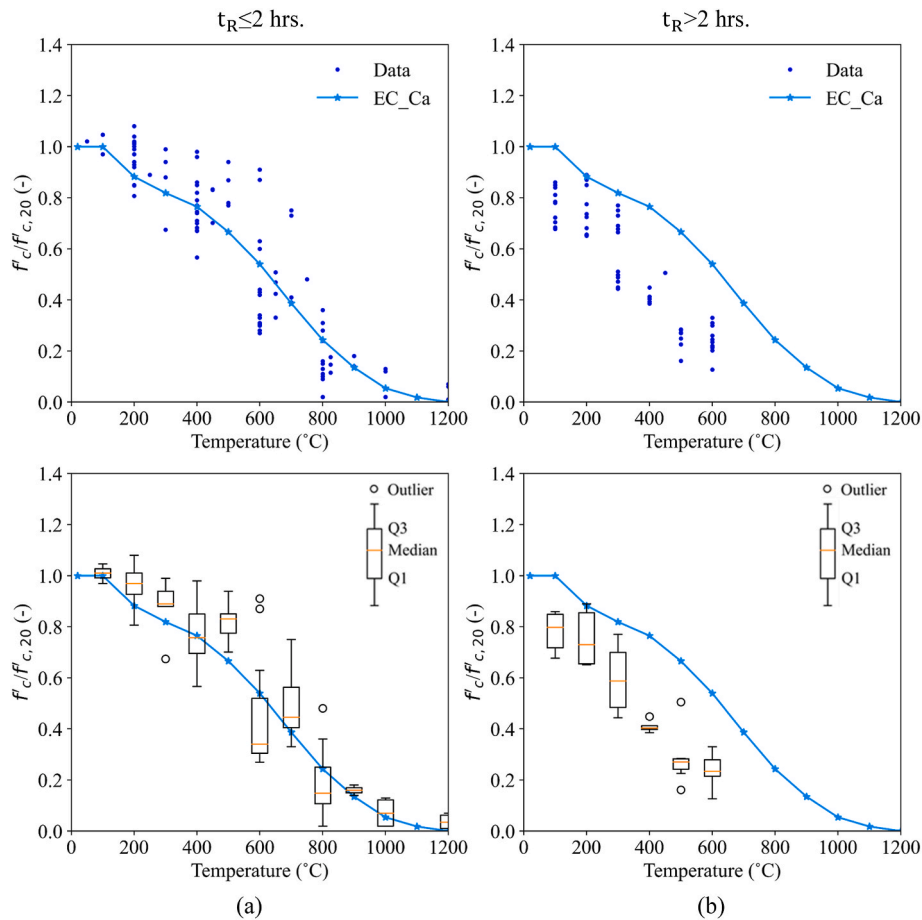


Fig. 11. Normalized residual strength of concrete with calcareous aggregate and retaining time (a) less than or equal to 2 h, (b) greater than to 2 h.

aggregates to confirm the conclusions.

Chen et al. [21] indicated that understanding the cooling process is essential for assessing the structural behavior of concrete exposed to fire as it affects the recovery of compressive strength. Only two studies, [22, 39], investigated the effect of the cooling rate, indicating that a faster cooling rate causes the concrete strength and stiffness to degrade more rapidly. A few studies [1,12,14,21,39,43,52,55,71] also looked at the effect of cooling type (air vs. water) for siliceous concrete. Fig. 12 plots the collected dataset for the residual compressive strength of concrete based on the cooling type. All of the data for calcareous aggregate were air-cooled. The majority of data for siliceous aggregate were also

air-cooled with some experiments using water. Most of the experiments reported that the relative strength of water-cooled specimens was lower than those of air-cooled specimens (e.g., [14,21,71]). However, the identified pattern is not significant when comparing all of the data-points; thus, the rest of this paper will not differentiate the cooling type in the data.

4.4. Discussion on the collected database

A few key observations can be made from the review and analysis of the collected dataset on the residual compressive strength of concrete

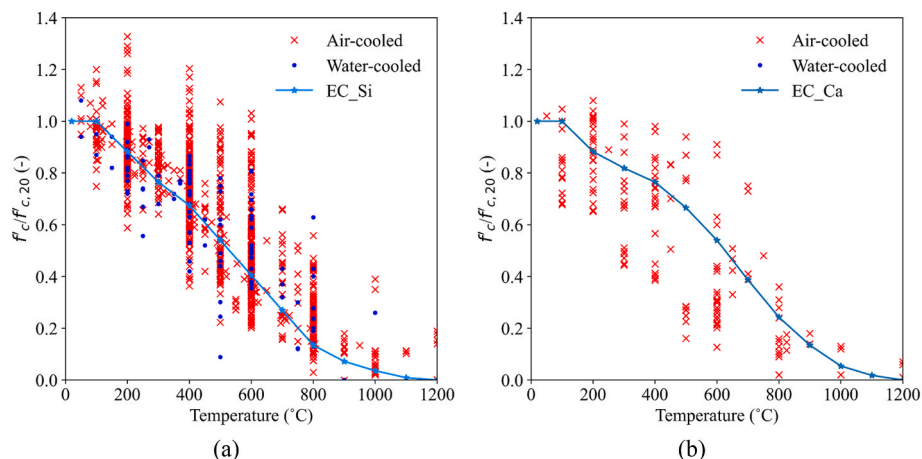


Fig. 12. Normalized compressive strength of concrete with (a) siliceous (b) calcareous aggregates based on cooling type.

subjected to high temperatures: (1) the majority of existing research has focused on concrete with siliceous aggregates, with limited studies targeting concrete with calcareous aggregate, (2) studies with variations in the concrete mix are dominant when compared to other attributes, stemming from the aim to investigate and improve the likelihood of spalling and/or cracking at high temperatures, and (3) more studies are needed to explore the effects of heating and cooling stages, and recommend a systematic and uniform testing protocol for high-temperature experiments. Given the variations in the attributes of the dataset and the need for additional research for a more systematic and uniform testing protocol, the scatter in the current dataset cannot be reduced much further after categorizing the data based on aggregate type (i.e., siliceous and calcareous) and the duration of retaining time (less than and greater than 2 h).

The next section discusses a probabilistic model based on the collected experimental data on the residual compressive strength of concrete and compares the results with the EC4 reduction factors. To this end, two models, one for concrete with siliceous aggregates and one with calcareous aggregates will be proposed. For concrete with siliceous aggregate, the refined dataset with a retaining time longer than 2 h will be used to build the probabilistic model. Given the limited number of data points for calcareous concrete, all collected data will be employed when building the model.

4.5. Probabilistic model for post-fire residual strength of concrete

This section will discuss and develop a probabilistic model for the residual compressive strength of concrete after exposure to high temperatures (i.e., post-fire) based on the collected and refined dataset in the previous sections. Qureshi et al. [50] quantified uncertainties in retained strengths of steel and concrete at high temperatures (i.e., during fire). Different distribution candidates were used with the collected data to identify the most suitable probabilistic model. In addition, the models were considered to be continuous functions that could be implemented in analytical and computational frameworks. A similar approach by Qureshi et al. [50] is used in this section to propose probabilistic models for post-fire residual compressive strength of concrete. The proposed models in this paper will complement the previously developed ones for analysis of structures during and after fires.

4.6. Evaluate candidate distributions

A total of 401 and 159 data points, collected from the literature, were used to develop the probabilistic models for residual compressive strength of normal strength concrete containing siliceous and calcareous

aggregates, respectively. As stated previously, the data points for concrete with siliceous aggregate are filtered to have a retaining time of more than 2 h to remove some of the large dispersions at certain temperature intervals (refer to Figs. 5a and 10). Raw data were then grouped at temperature intervals of 50 °C to capture variability in data at distinct temperature intervals. Each temperature group comprised data points within a ± 10 °C range, and datasets containing less than 5 data points were discarded. Consequently, 11 temperature groups for siliceous aggregate concrete and 7 temperature groups for calcareous aggregate concrete were created. Fig. 13 shows the raw data with dashed lines indicating the selected temperature groups, which will be used in the next step to develop the probabilistic models. The groups range between 50 °C and 800 °C. Noting that the data is normalized using the average measured value of concrete compressive strength at 20 °C; therefore, the variability of data at ambient temperature is minimal. To capture the scatter in concrete compressive strength at room temperature, and for consistency with existing models for ambient design, a mean value of 1.0 with a coefficient of variation (COV) of 0.12 is imposed at 20 °C when developing the probabilistic models. The selected value of the COV is based on the recommendation by Holický and Sýkora [34] for normal strength concrete compressive strength at 20 °C.

In the next step, 13 distinct probability density functions (PDF) were chosen as competing models, and their quality was assessed to select the best candidate. The selected functions that were fit to the data groups at each specific temperature are the: gamma, normal, lognormal, logistic, loglogistic, inverse Gaussian, Gumbel, generalized t-distributions, beta, Rician, Nakagami, Bimbaum-Saunders, and Weibull. To compare the distributions, three information criteria were used to quantify the relative goodness of each fit across all temperature groups, namely: (1) Bayesian information criterion or BIC; (2) Akaike information criterion or AIC; (3) Corrected AIC or AICc.

- The BIC calculates the maximum likelihood estimate, by adopting the negative measure of the fit as $-2 \times \log(\text{likelihood})$ and penalizing complexity as a positive $\log(N) \times k$, where N is the number of samples and k is the total number of model parameters [59].
- The AIC also applies the penalized likelihood criterion, but instead of $\log(N)$, it weighs k with a constant value of 2 [5].
- The AICc is equal to $AIC + \frac{2k^2 + 2k}{N - k - 1}$, introducing an adjustment by the inclusion of an additional penalty for small-sized datasets [35].

The three listed criteria were first computed at every temperature group for each of the 13 candidate distributions. Then, the mean across all the temperature groups was evaluated for each of the three criteria. Each bar in Fig. 14 represents the mean value for the three information

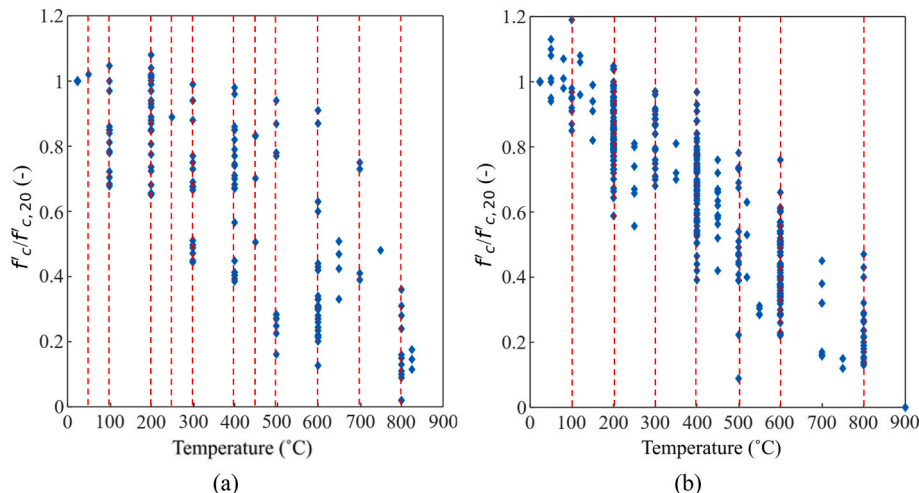


Fig. 13. Grouped data points for concrete with (a) siliceous (b) calcareous aggregates.

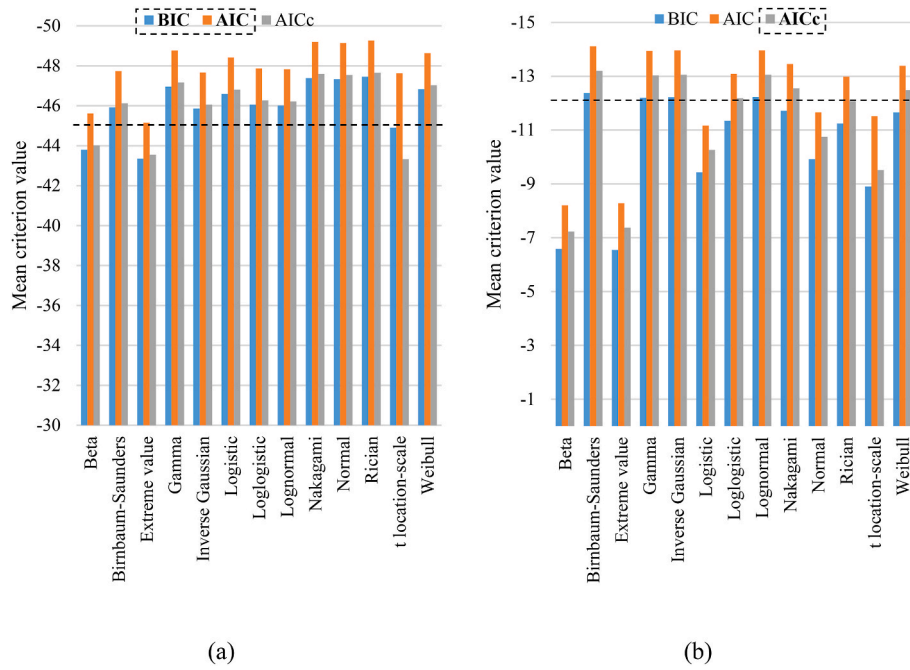


Fig. 14. Comparison of the mean criterion value for each of the 13 candidate PDF models for the residual compressive strength of concrete with (a) siliceous (401 points) and (b) calcareous (159 points) aggregates.

criteria and all candidate distributions. A model with a lower BIC, AIC, or AICc value is considered to be of higher quality (noting that the values in Fig. 14 are negative). The concrete data with calcareous aggregates had a small number of datapoints, for which the AIC may overfit whereas the BIC may underfit. Therefore, the AICc is a more suitable criterion to identify models with a better fit.

Fig. 14 shows that several distributions perform relatively better than the others considering both siliceous and calcareous datasets: Birnbaum-Saunders, gamma, inverse Gaussian, lognormal, Nakagami, and Weibull. A few additional requirements are imposed to select the

final stochastic functions:

- The chosen models should be a “continuous function” that appropriately captures scatter within experimental data for material strengths ranging from 20 °C to 1000 °C.
- There should be a “closed-form function” available for presenting quantiles of the model in order to facilitate implementation in analytical and computational frameworks.

On the basis of these additional requirements, the selected models

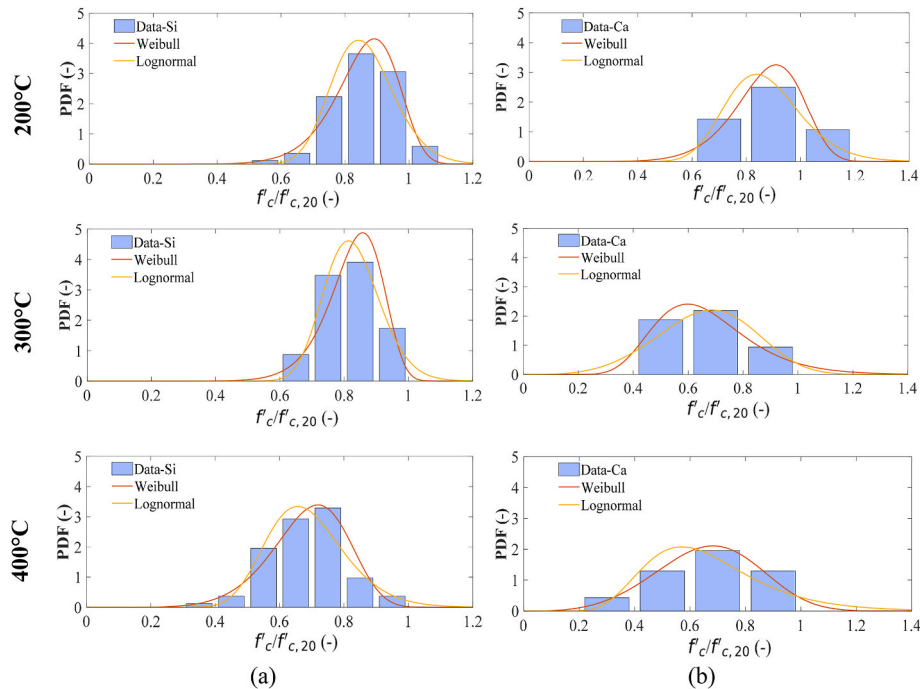


Fig. 15. Selected probabilistic models along with PDF for normalized residual strength for (a) siliceous and (b) calcareous concrete data at 200°C, 300°C, and 400°C.

were narrowed down to two distribution models: Weibull and lognormal. Fig. 15 shows the PDF for the Weibull and lognormal models against data for 200 °C, 300 °C, and 400 °C. Both distributions could represent the data; however, the Weibull function was chosen for both aggregate types moving forward. This ensures compatibility with established stochastic functions for concrete strength at high temperatures by Qureshi et al. [50] and prevents potential confusion in use of the models in future.

4.7. Temperature-dependent parameters of the Weibull distribution

The Weibull distribution is a PDF with two variables: scale (λ) and shape (k) parameters, as defined below:

$$f(x; \lambda, k) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-\left(\frac{x}{\lambda}\right)^k}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (1)$$

The two parameters of the Weibull distribution were determined for each temperature group of the concrete data with siliceous and calcareous aggregates, as presented in Fig. 16. Nineteen alternative polynomial functions were selected to fit to the parameters as a function of temperature. The R-squared value was calculated for each polynomial fit and used to assess how well the data matched the regression function. A larger R-squared value implies that the polynomial function provides a better match with the data. Aside from the R-squared value, a few other factors were considered to select the function with the best estimate. For example, the scale parameter (λ) should have a decreasing trend for the temperature range of 20–1000 °C, and the shape parameter (k) should be greater than 1. The red lines in Fig. 16 are the best fit polynomial regressions for the scale and shape parameters of the Weibull distributions for the residual strength of siliceous and calcareous concretes. The

following are the corresponding equations for the temperature-dependent parameters of the Weibull distribution for siliceous concrete:

$$\lambda(T) = 1.06 \times 10^{-3}T + 1.11 \quad (2)$$

$$k(T) = \frac{95.27}{5.095 + 23.21 \times 10^{-3}T} \quad (3)$$

Similarly, temperature-dependent parameters of the Weibull distribution for calcareous concrete can be determined using the following equations:

$$\lambda(T) = 1.06 \times 10^{-3}T + 1.068 \quad (4)$$

$$k(T) = \frac{18.92}{1.617 + 8.38 \times 10^{-3}T} \quad (5)$$

The temperature-dependent polynomial equations for λ and k are used with the Weibull distribution function to plot the 0.5, 0.05, and 0.95 quantiles as shown Fig. 17. The recommended EC4 reduction factors and experimental data from literature are also shown in the figure. It can be seen that the EC4 recommendation is relatively close to the median of the Weibull distribution for siliceous concrete, and the model captures the uncertainty in collected data from the literature. For calcareous concrete, the median of the Weibull distribution is lower than the EC4 recommendation. If a refined dataset with a retaining time greater than 2 h were used to develop the probabilistic model, the median would have been even lower. Thus, it appears that the EC4 model tends to overestimate the residual compressive strength of concrete with calcareous aggregate.

5. Summary and conclusion

This paper collected and analyzed experimental data on the residual

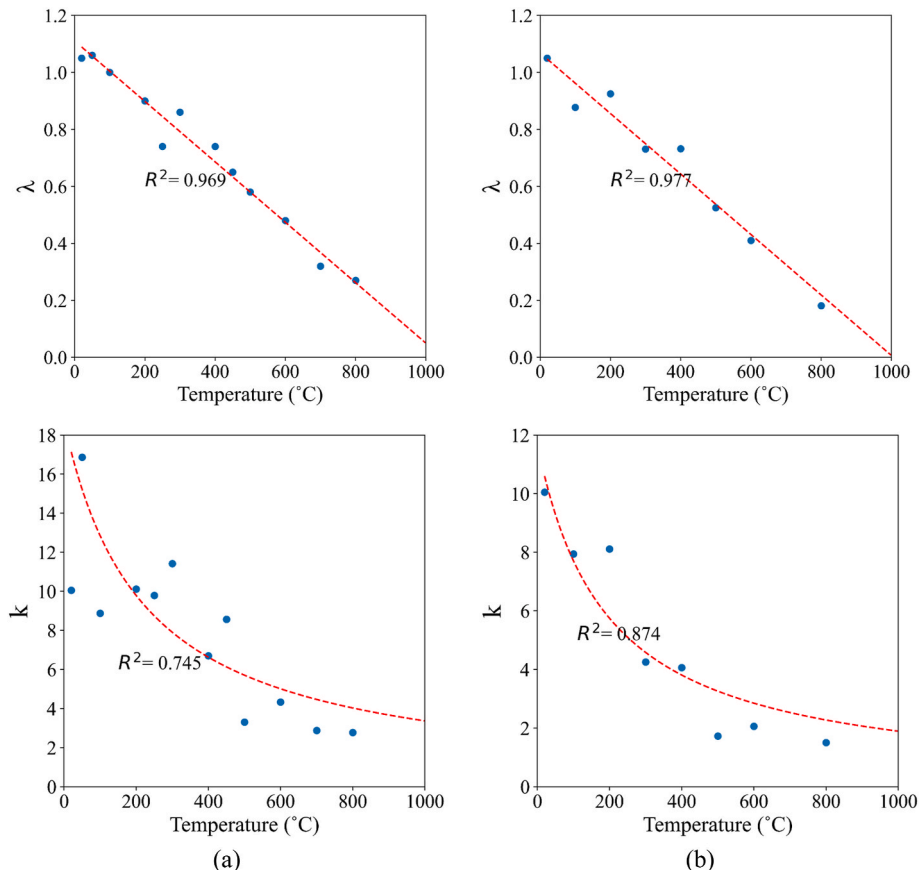


Fig. 16. Temperature-dependent polynomial regression for λ and k representing (a) siliceous concrete and (b) calcareous concrete.

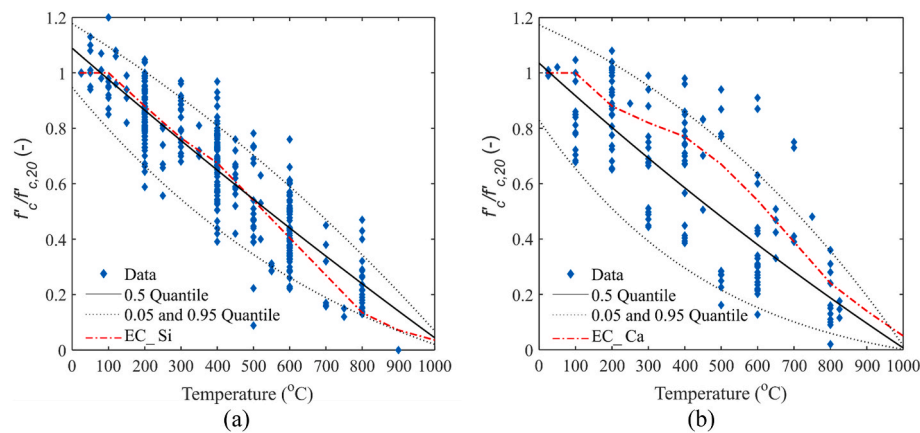


Fig. 17. Temperature-dependent residual strength data for normal strength concrete versus the Weibull probabilistic model and EC4 recommendation for: (a) siliceous concrete, and (b) calcareous concrete.

compressive strength of normal-strength concrete after exposure to high temperatures. A total of 53 papers, published after 1990, were reviewed and 1240 datapoints on the residual compressive strength of concrete as a function of temperature were collected. The dataset was first analyzed to identify attributes that could affect the residual strength of concrete. The refined dataset was then utilized to propose a probabilistic model for the residual compressive strength of concrete to capture the uncertainties in material strength as a function of the maximum exposure temperature. Conclusions of this study can be summarized as below:

- In the collected database, only 159 datapoints (of the 1240) studied concrete with calcareous aggregates. Thus, the majority of existing research has focused on siliceous concrete with limited data available for calcareous concrete.
- There is no standard testing protocol to evaluate degradation of mechanical properties of concrete at high temperatures. Thus, the employed testing protocol in existing experiments varied. Analyses of data showed that the duration of the retaining time at the maximum temperature had a significant influence on the scatter of the data. The specimen should be heated for enough time so that the core reaches the target temperature. Based on the results of this study, a minimum retaining time of 2 h is recommended to ensure uniform temperature in the specimen. Overall, there is a need to research a standard testing protocol that defines acceptable heating rates, retaining times, and cooling rates.
- Probabilistic models utilizing Weibull distributions were established to quantify uncertainties in the temperature-dependent residual compressive strength of siliceous and calcareous concrete. The parameters of the Weibull distributions were defined as continuous closed-form solutions as a function of temperature. Data points with a retaining time of at least 2 h were used to establish the probabilistic model for siliceous concrete, whereas the full dataset for calcareous concrete, including data with retaining times less than 2 h, was used to develop the model.
- The medians of data and proposed probabilistic model for the siliceous concrete are close to the EC4 recommendation (i.e., EC2 retention factors with an additional 10% reduction in strength). However, the EC4 recommendation tends to overestimate the reduction factors for calcareous concrete when compared with the medians of the data and the proposed model.

Author statement

Marzieh Shahraki: Conceptualization, Methodology, Data curation, Formal analysis, Writing - original draft, Writing - review & editing, Nan Hua: Conceptualization, Methodology, Data curation, Writing - review

& editing, Negar Elhami-Khorasani: Conceptualization, Methodology, Writing - review & editing, Supervision, Funding acquisition, Anthony Tessari: Conceptualization, Methodology, Writing - review & editing, Supervision, Funding acquisition, Maria Garlock: Conceptualization, Methodology, Writing - review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the CAIT Region 2 UTC Consortium, and the Institute of Bridge Engineering (IBE) at the University at Buffalo. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the CAIT Region 2 UTC Consortium.

The authors are grateful for the work of Krystal Delonce and Jessica Cui, undergraduate students at Princeton University, who collected the data for this research.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.firesaf.2022.103698>.

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